

- 9 (a) Radioactive decay is a random and spontaneous process.

Explain what is meant by

- (i) a *random* process,

It is a process in which it cannot be predicted which nuclide will decay at a particular instant or when any particular nuclide will decay.

[1]

- (ii) a *spontaneous* process.

It is a process which it is not triggered or affected by external factors such as temperature and pressure.

[1]

- (b) An unstable nucleus of mass number A undergoes α -decay, as illustrated in Fig. 9.1.

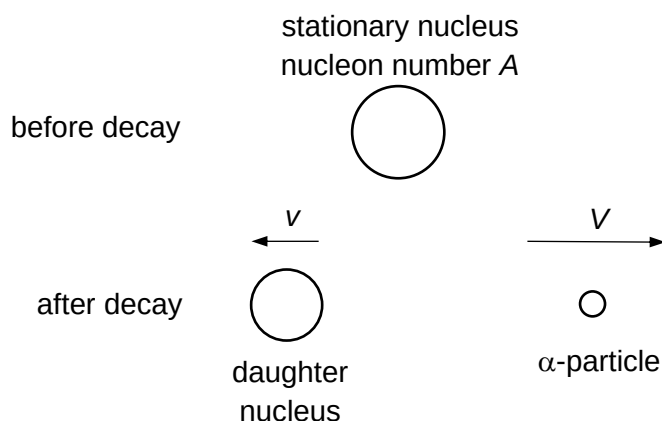


Fig. 9.1

The nucleus is stationary before the decay.

After the decay, the initial speed of the α -particle is V and that of the daughter nucleus is v .

- (i) Derive an equation, in terms of A , v and V , to represent conservation of linear momentum for this decay.

[1]

$$\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$$

$$0 = 4V + (A-4)v$$

$$\frac{V}{v} = \frac{A-4}{4} \quad [1]$$

(ii) Show that the ratio

$$\frac{\text{initial kinetic energy of } \alpha - \text{particle}}{\text{initial kinetic energy of daughter nucleus}}$$

is equal to $(\frac{1}{4}A - 1)$.

$$\begin{aligned} \frac{\text{initial KE of } \alpha\text{-particle}}{\text{initial KE of daughter nucleus}} &= \frac{\frac{1}{2}(4)v^2}{\frac{1}{2}(A-4)V^2} && [1] \text{ ratio} \\ &= \frac{4}{A-4} \left(\frac{v}{V} \right)^2 \\ &= \frac{4}{A-4} \left(\frac{A-4}{4} \right)^2 \\ &= \frac{A-4}{4} && [1] \text{ subst} \\ &= \frac{1}{4}A - 1 \end{aligned}$$

[2]

(c) Data for the α -decay of bismuth-212 ($^{212}_{83}\text{Bi}$) to form thallium-208 ($^{208}_{81}\text{Tl}$) are given in Fig. 9.2.

nucleus	mass of nucleus / u
bismuth-212	211.9459
thallium-208	207.9374
helium-4	4.0015

Fig. 9.2

(i) Use the data of Fig. 9.2 to calculate, to two places of decimals, the energy released during the decay.

$$\begin{aligned} \text{total rest mass of reactants} &= 211.9459 \text{ u} && [1] \text{ mass diff/u} \\ \text{total rest mass} &= 207.9374 \text{ u} + 4.0015 \text{ u} && [1] \text{ to kg} \\ \text{mass difference} &= 0.007 \text{ u} && [1] \text{ to J} \\ \text{energy released/J} &= 0.007 \times (1.66 \times 10^{-27}) \times (3.0 \times 10^8)^2 && [1] \text{ to MeV} \\ \text{energy released/MeV} &= 0.007 \times (1.66 \times 10^{-27}) \times (3.0 \times 10^8)^2 / (1.6 \times 10^{-19}) = 6.54 \end{aligned}$$

$$\text{energy} = \dots\dots\dots \text{MeV} [4]$$

- (ii) Use your answer in (c)(i) to show that, based on the expression in (b)(ii), the energy of the α -particle is 6.42 MeV.

$$\frac{\text{initial KE of } \alpha\text{-particle}}{\text{initial KE of Tl}} = \frac{1}{4}(212) - 1$$

$$\frac{\text{initial KE of } \alpha\text{-particle}}{\text{total initial KE} - \text{initial KE of } \alpha\text{-particle}} = 53$$

$$\begin{aligned}\text{initial KE of } \alpha\text{-particle} &= 53 (\text{total initial KE} - \text{initial KE of } \alpha\text{-particle}) \\ &= 53 (6.54 - \text{initial KE of } \alpha\text{-particle}) \\ &= 6.42 \text{ MeV}\end{aligned}$$

[2]

- (d) In practice, the α -particle is found to have an energy of 6.10 MeV, rather than 6.42 MeV, as calculated in (c)(ii).

Suggest

- (i) an explanation for the difference in energy,

The energy released in a nuclear reaction are in the form of the kinetic energy of the product particles or energy of an emitted γ photon. Since KE of the product particles is smaller than the total energy lost in the reaction, this implies there is an emission of γ -photon(s).

[1]

- (ii) why it is likely that the thallium nucleus and the α -particle do not move off in opposite directions.

Since there is emission of gamma photon(s), the total momentum of zero would include the momentum of the gamma photon(s) as well as the momentum of the thallium nucleus and that of the α -particle. The thallium nucleus and particles need not move off directly in opposite, they could move in other directions such that the vector sum of momentum of the thallium nucleus, α -particle and gamma photon(s) is zero.

[2]

- (e) Some data for the half-lives and decay constants of bismuth-212 and thallium-208 are given in Fig. 9.3.

nucleus	half-life / s	decay constant / s^{-1}
bismuth-212		1.9×10^{-4}
thallium-208	190	3.7×10^{-3}

Fig. 9.3

- (i) Define *half-life*.

..... The half life $t_{1/2}$ of a quantity is defined as the time taken for the quantity to decrease to half of its initial value. [1]

- (ii) Complete Fig. 9.3 by calculating the half-life of bismuth-212.

$$T_{1/2} \text{ of bismuth 212} = 0.693 / \square = 0.693 / 1.93 \times 10^{-4}$$

[1]

- (iii) Initially, a radioactive source contains N nuclei of bismuth-212.

After two hours, it is found that the number of bismuth-212 nuclei has reduced to approximately $\frac{1}{4} N$. However, although bismuth-212 decays to form thallium-208, the number of thallium nuclei is much less than $\frac{3}{4} N$.

Suggest an explanation for these observations.

Based on the data given, the half life for bismuth 212 is 3700 s while the half life for thallium 208 is only 190 s. Hence half life for bismuth 212 is significantly larger than half life for thallium 208.

Hence in the first 190 s of decay, the number of bismuth-212 will remain approximately the same while the number of thallium 208 will have reduced by half.

After two hours, bismuth -212 would have gone through about 2 half lives, hence the number of bismuth 212 is about $N/4$. But thallium 208 is not a

stable nucleus, hence the number of thallium after two hours must be less than $3/4 N$. [Total: 20]

Since thallium 208 only have a half life of 190 s, after 2 hours, the number of thallium would have reduced greatly and hence much less than $3/4 N$.

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